



**SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)**

Constituent of Symbiosis International (Deemed University), Pune

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**Subject: Programming with Java**

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**Div: CSE A1**

**Batch:2025-29**

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### **ASSIGNMENT No : 1**

**TITLE :**Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

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### **Problem Statement**

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

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### **Learning Objectives**

Understand the theoretical concepts, implement them in Java programs, analyze outputs, and apply the concepts to solve practical programming problems.

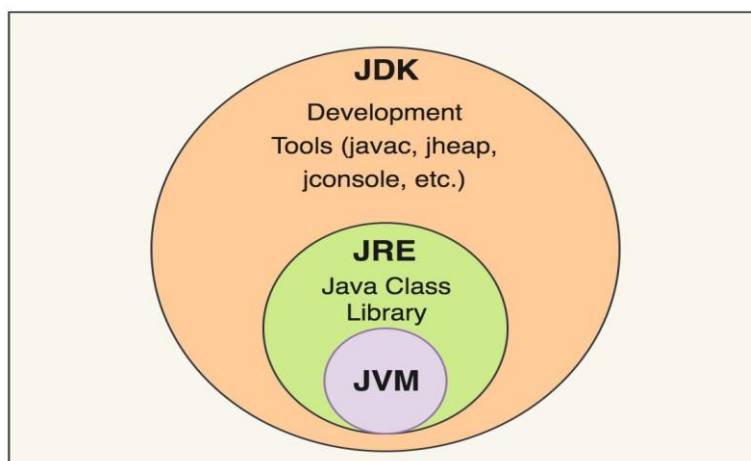
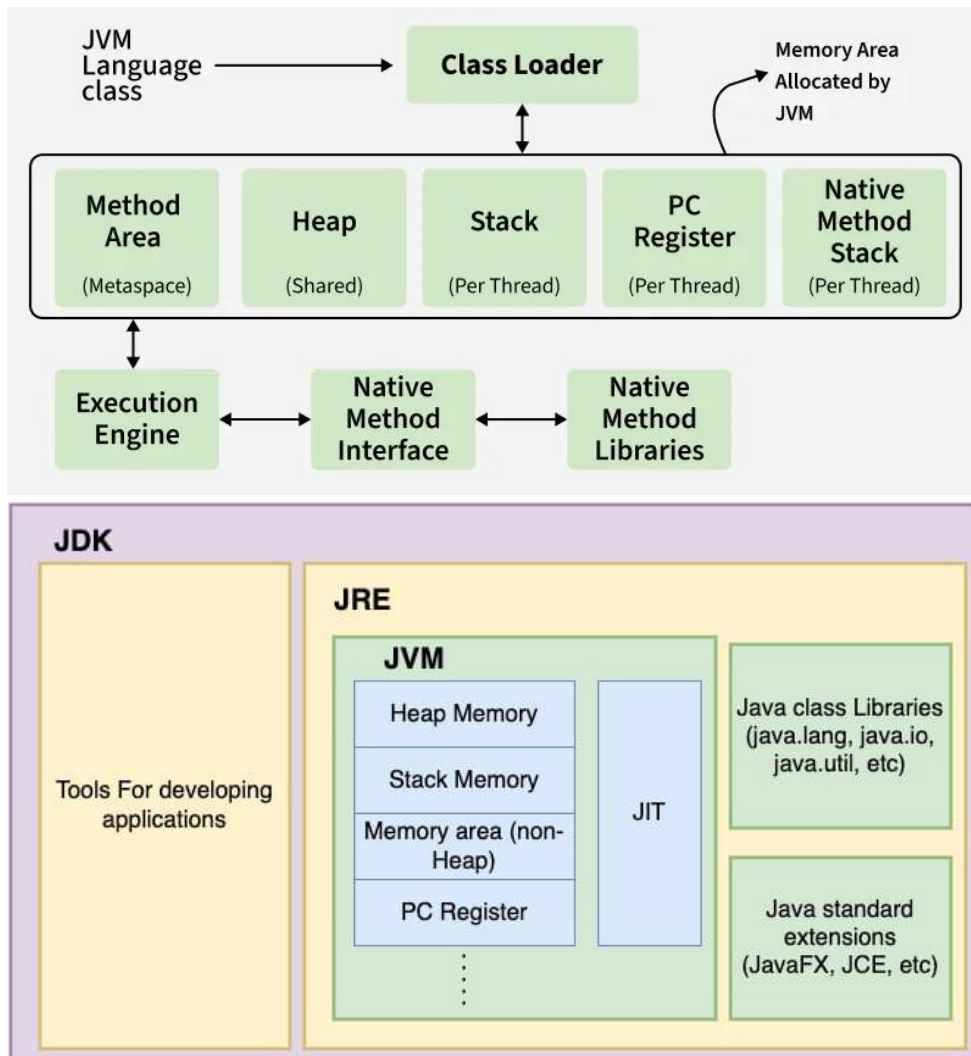
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### **Theory**

Java environment setup, JDK/JRE/JVM, variables, methods and constructors. forms an important component of Java programming. Students should understand the underlying concepts, architecture, advantages, use cases, and implementation methodology. The exercise should focus on understanding the design principles, coding practices, execution flow, and practical applications of these concepts in software development. Detailed study should include syntax, APIs, execution process, and real-world applications.

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## Output Screenshot:



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## Conclusion:

Thus, we have successfully understood the Java environment setup along with JDK, JRE, and JVM, and implemented the concepts of variables, methods, and constructors through suitable Java programs. The programs were compiled and executed successfully, and the outputs were verified as shown in the screenshots.

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## Questions

### 1. Explain the main concept used in Java environment setup, variables and methods.

Java environment setup means installing the JDK on your system and setting the PATH so that the javac (compiler) and java (runtime) commands can be used from the terminal/command prompt. Once installed, the JDK gives us tools to write, compile, and run Java programs, while the JVM inside the JRE actually executes the compiled .class bytecode.

Variables are named memory locations used to store data (like int, double, String) that a program can use and change during execution. Methods are blocks of code written inside a class that perform a specific task — they take input (parameters), process it, and can return a result. Together, variables hold the data and methods define the operations performed on that data, which is the core idea behind writing any Java program.

### 2. What are the advantages of using Java variables and methods?

Variables: allow data to be stored and reused instead of hardcoding values, make programs flexible (values can change at runtime), and improve readability by giving meaningful names to data.

Methods: allow code reusability (write once, call many times), make programs easier to debug and maintain by breaking logic into smaller units, support modularity (each method does one job), and reduce code duplication across a program.

### 3. How can this implementation be improved?

### 4. What are the real-world applications?

- Environment setup (JDK/JRE/JVM) is the foundation for all Java-based software — enterprise applications, Android apps, web servers, and desktop tools all rely on it.
- Variables and methods are used everywhere in real applications: e.g., in a banking app, variables store account balance and customer details, while methods handle operations like deposit(), withdraw(), or checkBalance().

- Constructors are used to initialize objects automatically when they are created — for example, creating a new Employee object with its ID and name already set.
- These concepts together form the base for larger real-world systems like billing software, inventory management, mobile apps, and web applications.